



The bridge to possible

SRv6 for 6G

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Motivation

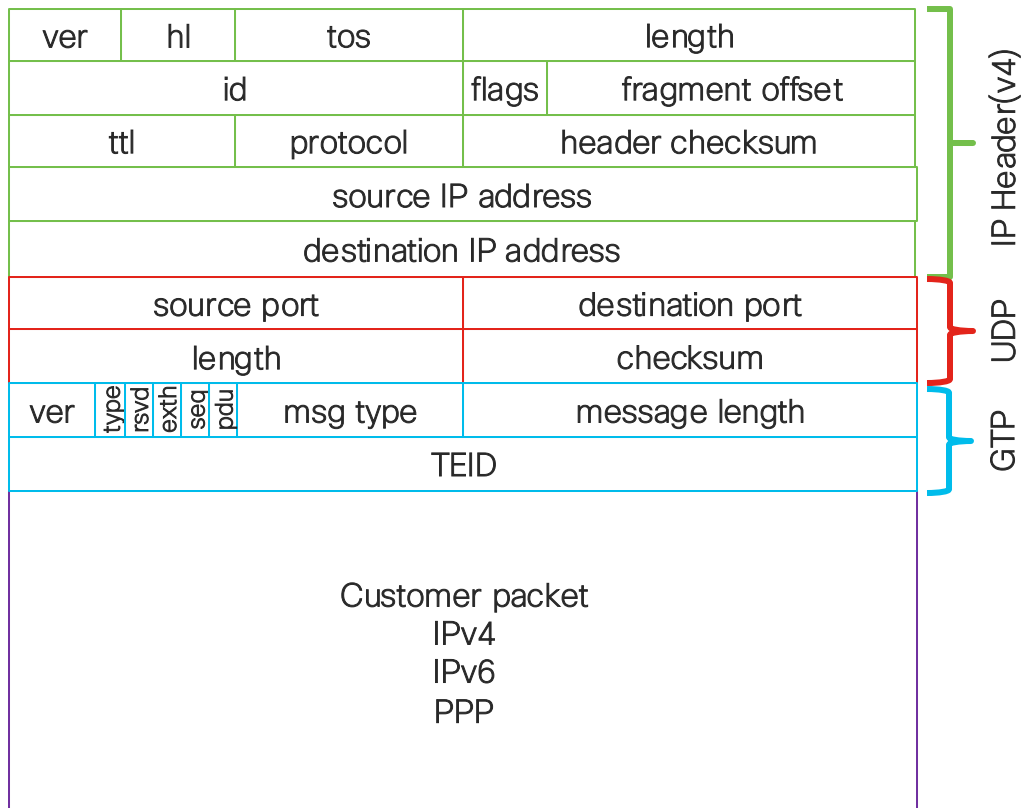
- GPRS Tunneling Protocol (GTP) faces scalability, security, and efficiency challenges in 6G and beyond networks, limiting support for ultra-low latency and advanced applications.
- Segment Routing over IPv6 (SRv6) offers a more flexible, scalable, and programmable alternative, eliminating tunneling overhead and enhancing traffic engineering and network slicing.

GTP (GPRS Tunneling Protocol)

- Designed in 90s (release 1998)
- Originally for SGSN-GGSN communication only
- Essentially two protocols GTP-C and GTP-U
- GTP-C – used to establish PDP context
- GTP-U – used to carry customer data (IPv4, IPv6, PPP)
- GTP' – another flavor used for charging only
- In 5G GTP-U is used on N3,N2,N9,N4 interfaces for dataplane

GTP-U packet

- Outer IP header used for address space separation
- UDP potentially used for entropy encoding (not used)
- GTP header carries TEID – to identify PDU session
- In 5G QFI extension header is mandatory



GTP Limitations

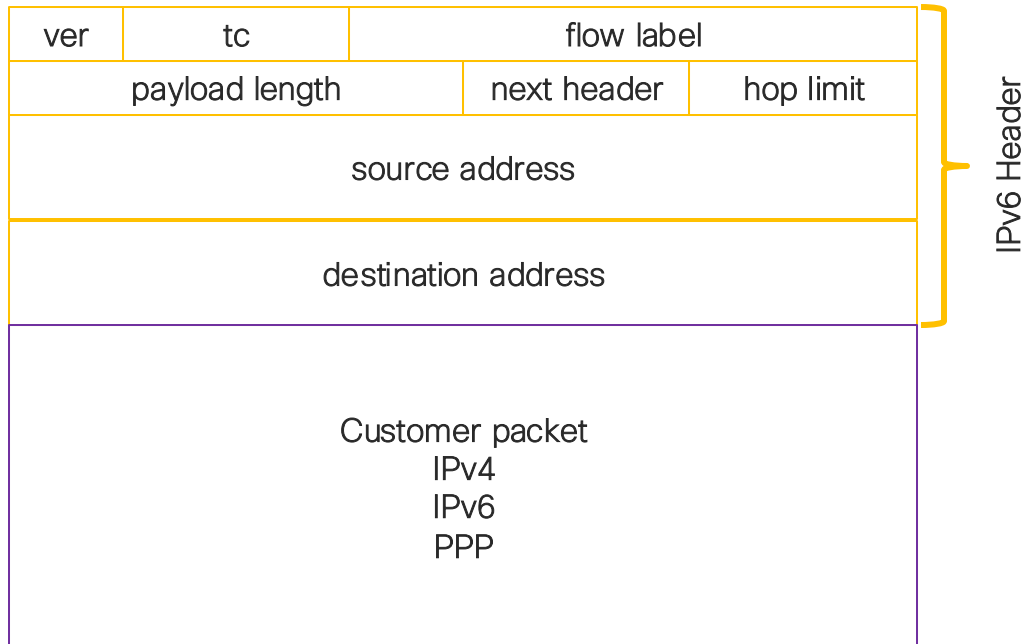
- Does not allow to define data any SLA and properties for IP transport network (except DSCP)
- Lack entropy of inner packet
 - TEID needs to be included in hashing -> complex transport HW
- Significant overhead (IP+UDP+GTP)
- GTP is stateful

SRv6

- Modern Universal Protocol
- Plain IPv6 Encapsulation
- Extends Semantics of Destination IP – SID
- Allows encoding of specific path in transport network
- Allows to encode additional flexible metadata
- IPv6 Natively supports entropy encoding

SRv6

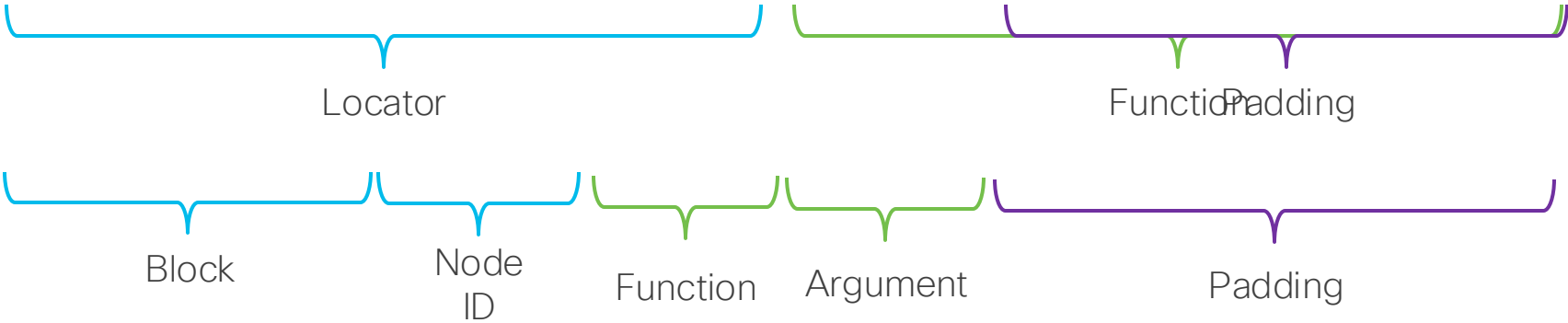
- IPv6 Header
- Destination IP address
 - Segment ID (SID)
- Flow Label = entropy of encapsulated packet



SID Structure

128 Bits Like IPv6 address but different semantics

1111:2222:3333:4444:5555:6666:7777:8888



SID For Mobility Example

1111:2222:3333:4444:5555:6666:7777:8888

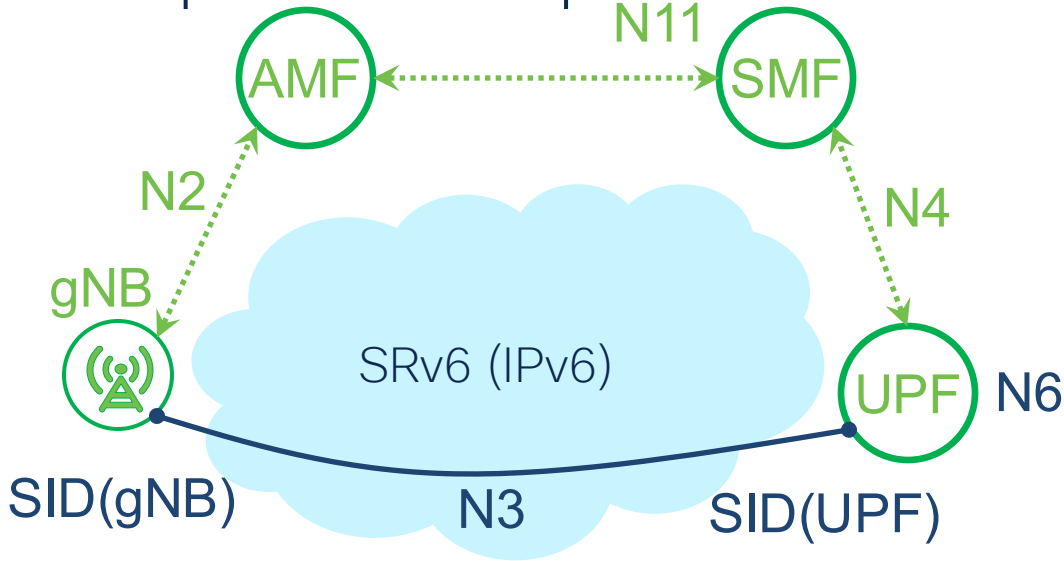


- Locator – assigned to gNB and UPF
- Function – Defines APN
- Argument 1 – TEID
- Argument 2 – QFI

Usecases

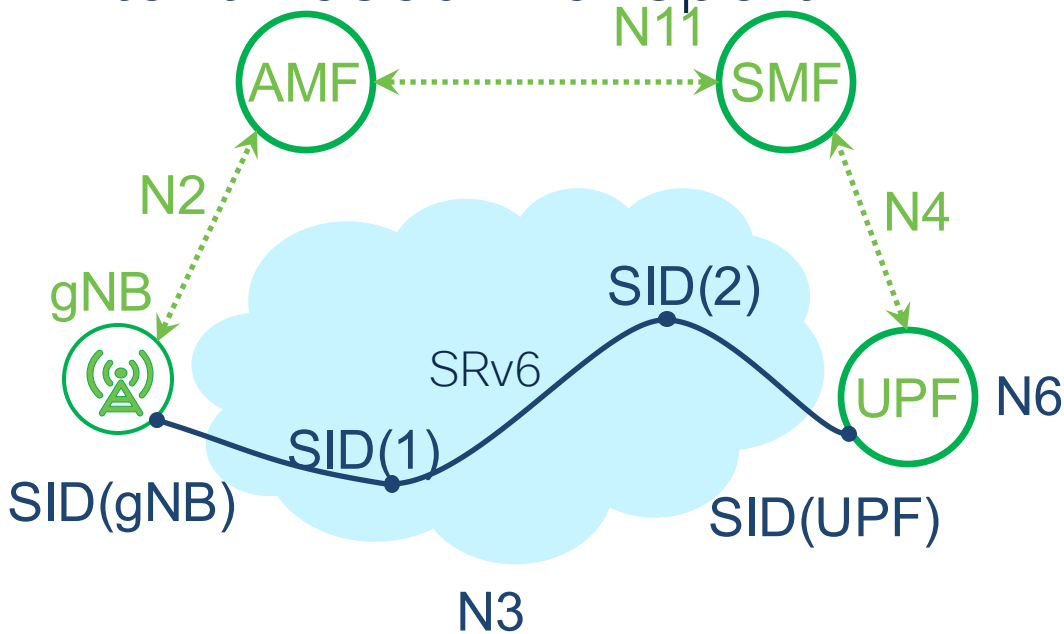
- Simple Replacement
- **Intent Based Forwarding**
- Low Latency
- Service Chaining
- **UPF Disaggregation**
- Replication
- Stateless UPF
- Anchorless UPF
- UPF less service
- Native Entropy
- Slice ID

Simple GTP Replacement



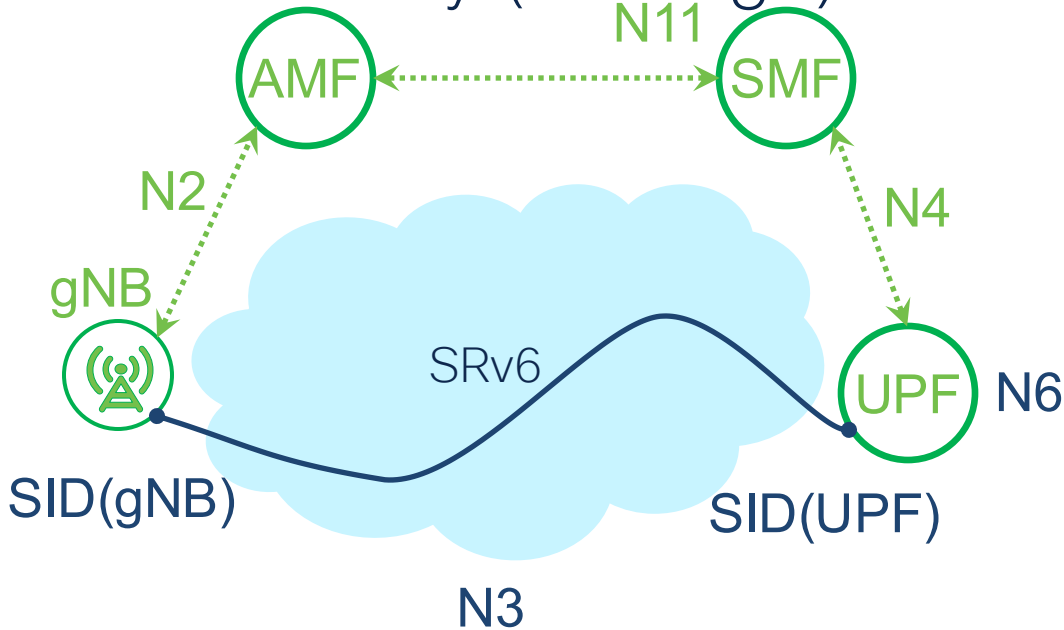
- Simple GTP replacement
- SID(gNB) has TEID and QFI arguments
- SID(UPF) has TEID and QFI arguments
- Backhaul network – Simple IPv6 routing

Intent Based Transport



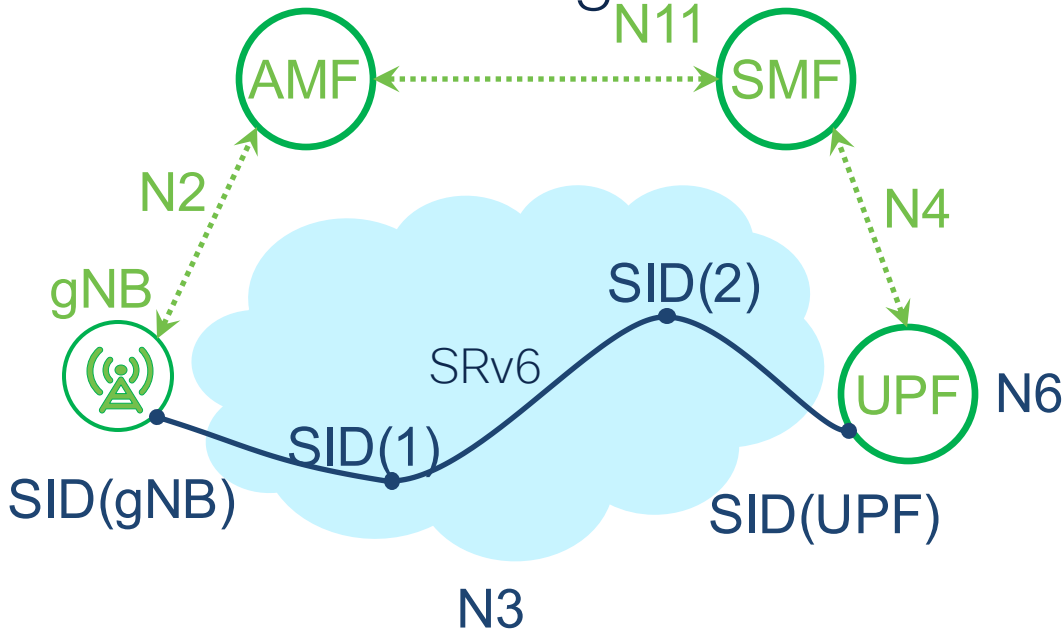
- UPF and gNB are programmed with multiple SIDs per UE to ensure constrained transport across Backhaul network
- Backhaul topology must be shared computed centrally from network topology (BGP-LS)
- Can be used for Low Latency

Low Latency (FlexAlgo)



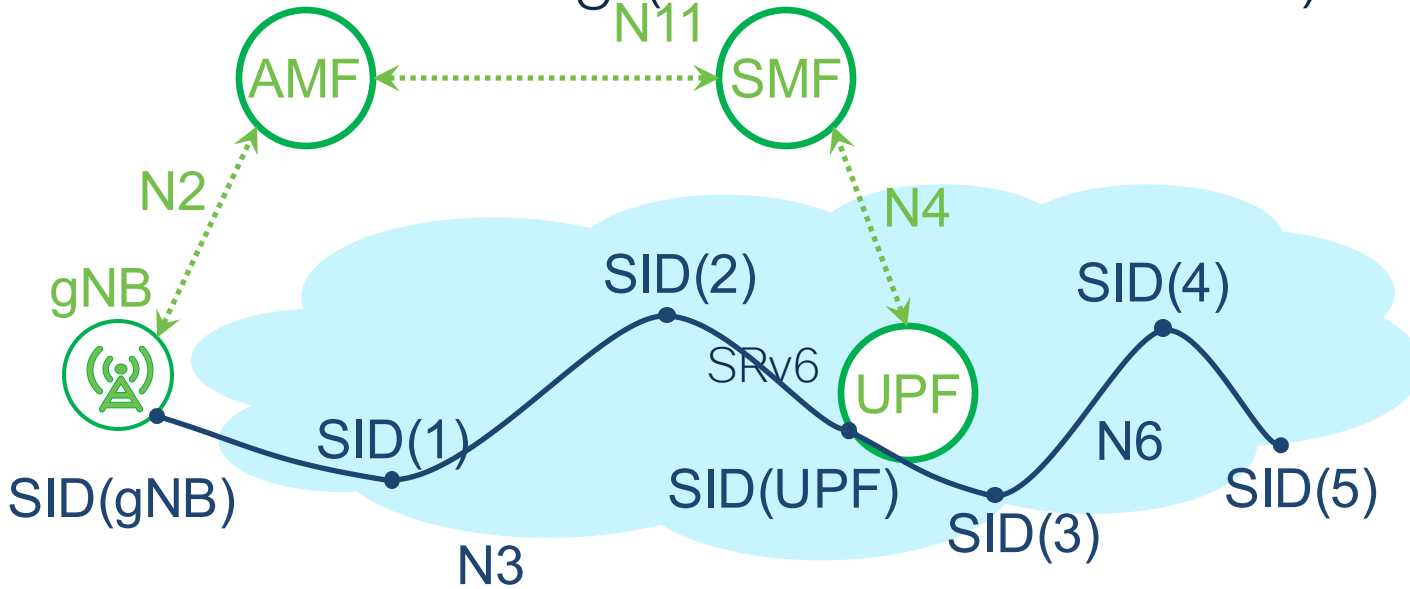
- UPF and gNB programmed single SID (with TEID+QFI)
- SID in the network is from Flexible Algorithm following Latency metric
- Network computes LL path
- Slicing in transport network

Service Chaining



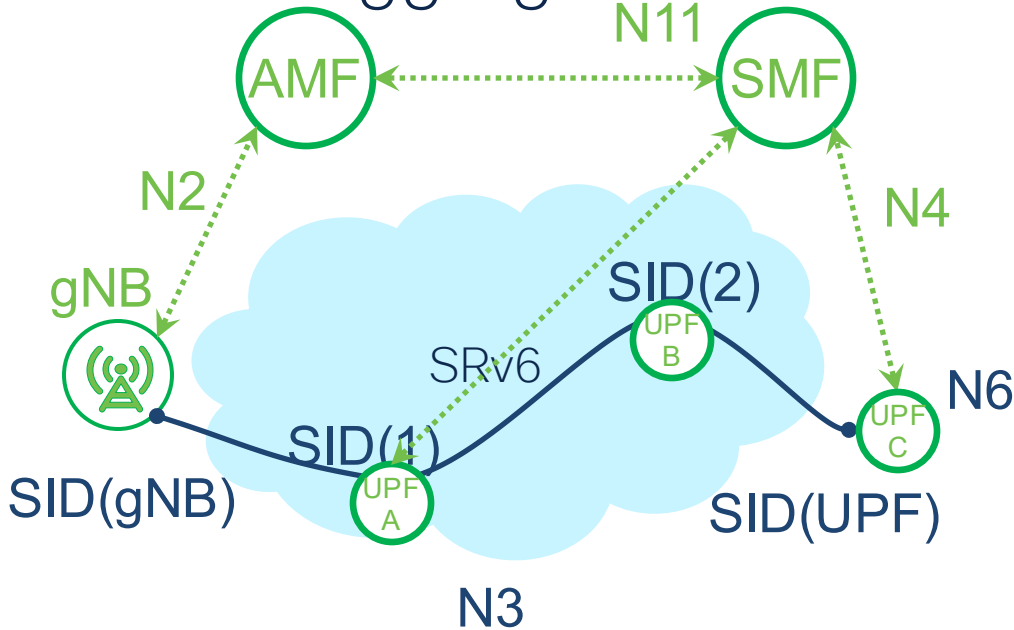
- UPF and gNB are programmed with multiple SIDs per UE to ensure that specific session is passing any required network functions.
- i.e Content Enrichment, Firewall IPS.....
- Per Session Services

Service Chaining (extended to N6-Gi)



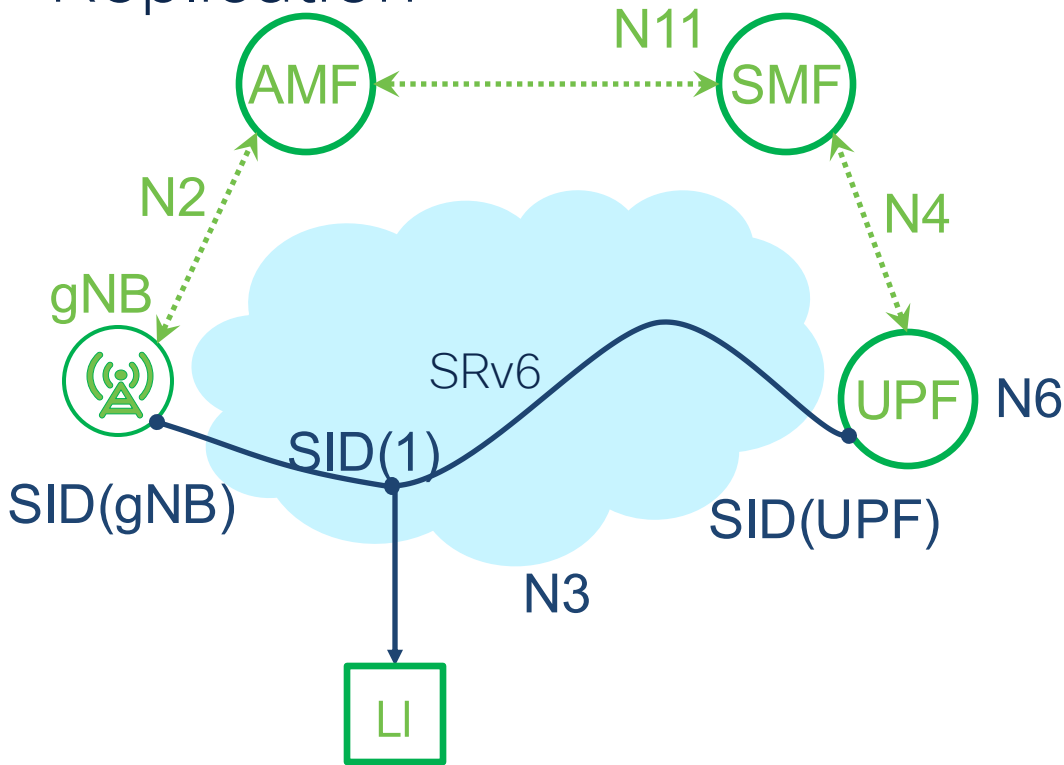
- UPF functionality can be performed anywhere in a Service Program
- Per session services

UPF Disaggregation



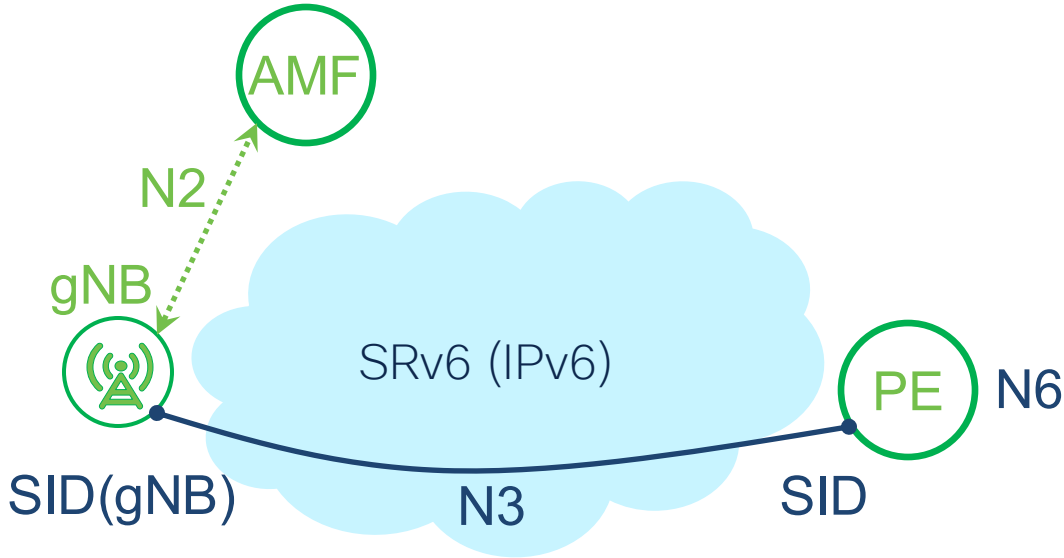
- UPF can be disaggregated into logical microservices
- Each microservice provides subset of UPF functions (QOS, Accounting, encap/decap, anchoring....)
- Each session then uses only relevant microservices
 - PDU1 (full mobility) goes UPF A, UPF B, UPF C
 - PDU2 (FWA) goes straight to UPF C
- Each UPF can be very optimized for it's function (HW, SW, VNF, CNF....)

Replication



- Specific SID Replicates selected traffic
- Typical use case LI or troubleshooting
- Replicates based on TEID
- 100% transparent

UPF Less



- Not all sessions needs full UPF functionality
- Session can be terminated directly in N6
- i.e Fixed Wireless
- Address mobility (anchoring per gNB)

Current 3GPP Status



- 4 proposal at CT4 – overall responsibility for Core network protocols
 - SRv6 Baseline Architecture
 - SRv6 Multi-hop forwarding Architecture
 - SRv6 Roaming Architecture
 - Interworking Architecture
- Cosigned by DT, Verizon, Vodafone, Sony, Apple
- Accepted proposals
 - Further evaluation
 - Other UP protocol options – QUIC, Enhanced GTP
- Opposed by Nokia, Ericsson, Huawei
- Next Meeting Oct 12-16 Prague **JOIN US**
- Any support is more than Welcome



The bridge to possible